

Write code logic for the given Scenarios:

1. A healthcare startup wants to build a model to predict the onset of diabetes. The dataset has 50 features, but the team wants to use only the most relevant ones. How can they apply feature selection effectively?

Answer:

Apply feature selection techniques such as RFE or SelectKBest. For example, using RFE with a Random Forest classifier, we can recursively remove less important features and select the top 10 features. Then we train the diabetes prediction model using only those selected features. PCA can also reduce dimensionality, but it is feature extraction rather than feature selection.

Step 1: Split features and target

Step 2: Split the dataset

Step 3: Create a model

Step 4: Apply RFE or SelectKBest to select important features

Step 5: Fit RFE/ SelectKBest

Step 6: Train the model using selected features

Step 7: Predict

Step 8: Evaluate

Display selected features

2. You're creating a new project for managing student data. The project name is student_portal. What is the step-by-step command to start it? write logic for this django project

Answer:

1. Install Django

```
pip install django
```

2. Create a new Django project

```
django-admin startproject student_portal
```

3. Move into the project folder

```
cd student_portal
```

4. Start the Django development server

```
python manage.py runserver
```

- After running the last command, Django usually gives a local address such as:
`http://127.0.0.1:8000/`

3. A company is analyzing customer purchase patterns with 200+ behavioral features. How can they reduce dimensionality without losing predictive power?

Answer:

- Since there are more than 200 features, Use PCA to reduce the number of features.
- PCA combines the information from the original features into fewer new components.
- Keep enough components to retain most of the information, such as 95% of the variance.
- Then train the model using these components and check whether the prediction performance is still good.

4. A digital library wants to recommend books to readers based on what similar readers liked. How should they design this system?

Answer:

- Build a user-item interaction matrix using users' book ratings or reading history.
- Calculate similarity between users using Cosine Similarity to find readers with similar interests.
- Find books liked by similar users that the target user has not read yet.
- Recommend those books to the target user.
- Handle missing ratings/sparse data using techniques such as KNN or matrix factorization.
- Evaluate the recommendation system using metrics such as Recall, Precision, or RMSE.
- Update recommendations whenever users provide new ratings or read new books.

5. A bank wants to assess the risk level of credit applicants using only the most important financial indicators. How can they reduce the number of features?

Answer:

- Start with all financial features such as income, credit score, debt, loan amount, and employment duration.
 - Use a tree-based model like Random Forest to calculate feature importance.
 - Remove less important or irrelevant features and keep the most important financial indicators.
 - Train a classifier using the selected features and evaluate its performance.
 - Evaluate the model using ROC-AUC, Precision, and Recall.
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6. A news app wants to recommend articles based on both article similarity and user reading history. How can they implement a hybrid system?

Answer:

- Collect article content and user reading history.
 - Use Content-Based Filtering to find similar articles based on article content.
 - Use Collaborative Filtering to analyze users' reading history and preferences.
 - Combine both recommendation results to create a hybrid recommendation.
 - Rank the recommended articles and show the most relevant ones to the user.
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7.Question:

You're building a spam detection model and have thousands of text features from emails. How do you identify the most useful ones?

Answer:

- Convert email text into numerical features using techniques such as TF-IDF or Bag-of-Words.
- Identify important features using methods such as SelectKBest, RFE, or model-based feature importance.
- Remove less useful features and keep the features that are most useful for detecting spam.

- Train a classification model such as Logistic Regression, Naive Bayes, or Random Forest using the selected features.
 - Evaluate the model using Precision, Recall, and F1 Score.
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8. An ed-tech platform wants to recommend courses based on what similar learners have enrolled in. What steps would you take?

Answer:

- Collect learner-course interaction data, such as enrollments, ratings, or course completion.
 - Find **similar learners** using methods like **KNN** or **Matrix Factorization (SVD)**.
 - Identify courses that similar learners have enrolled in or liked.
 - Recommend those courses to the target learner.
 - Evaluate the recommendation system using **Precision, Recall, and F1 Score**.
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9. You're developing a car price prediction tool. With 100+ features (e.g., brand, mileage, engine type), how do you reduce complexity?

Answer:

- Reduce complexity by removing redundant features using correlation analysis.
 - Selecting important features using RFE, SelectKBest, or feature importance from Random Forest/Gradient Boosting.
 - Since car price is a numerical output, this is a supervised regression problem.
 - Evaluate it using RMSE and R^2
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10. How do you recommend products to new users who haven't interacted with anything yet?

Answer:

- Recommend popular and trending products based on ratings, reviews, and engagement.
- Use content-based filtering if user interests or preferences are available.
- Gradually introduce collaborative filtering as the user starts interacting with products.
- Provide personalized recommendations based on the user's behavior.
- Evaluate recommendations using CTR, Precision, Recall, and engagement.

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